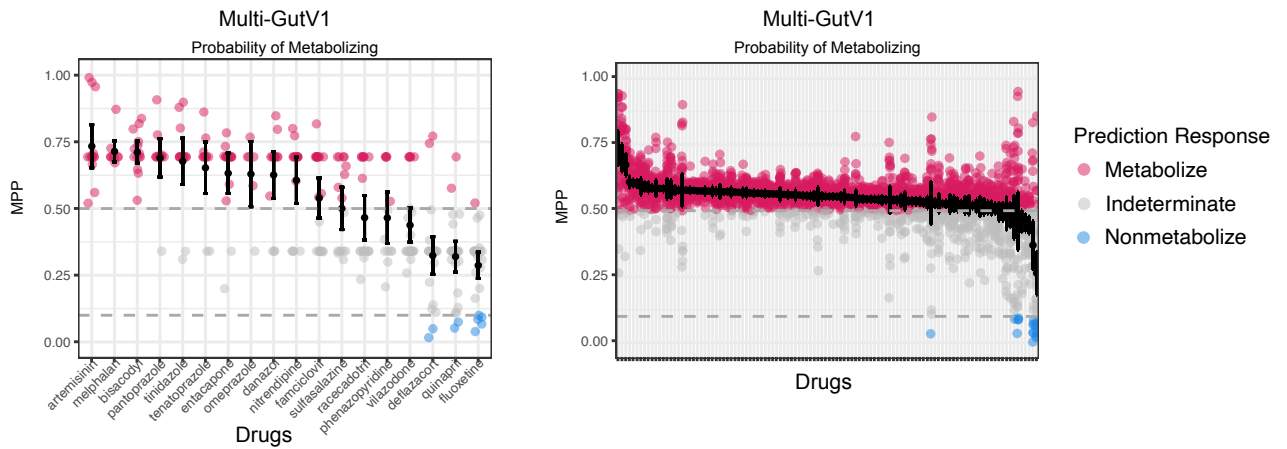
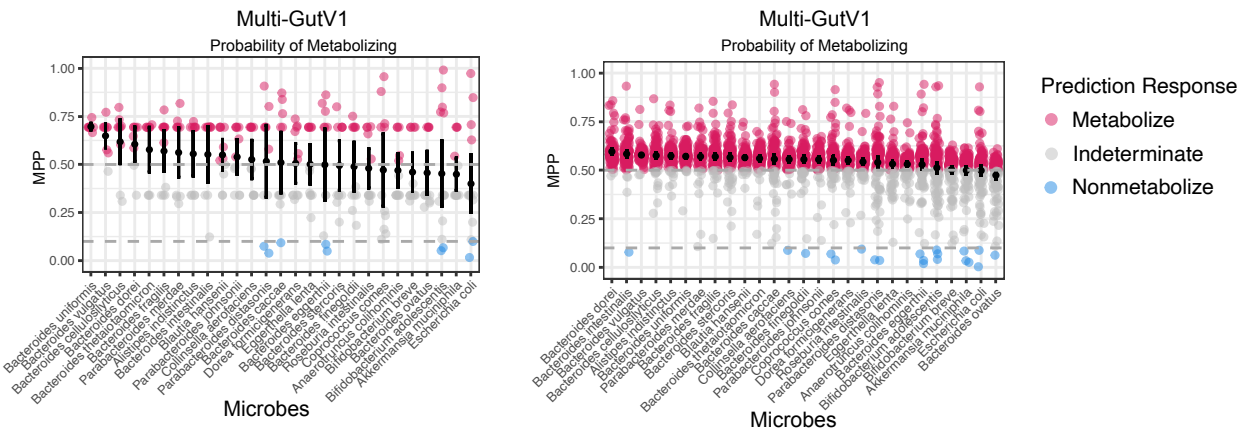
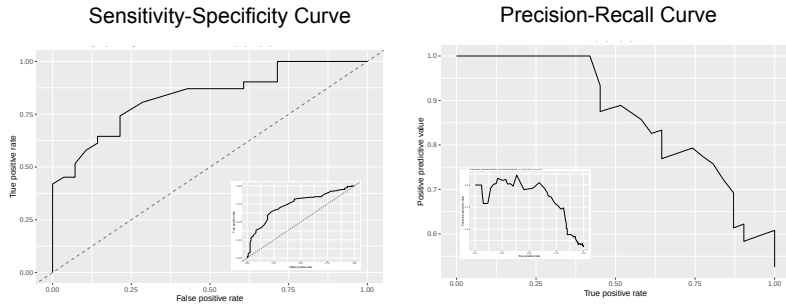
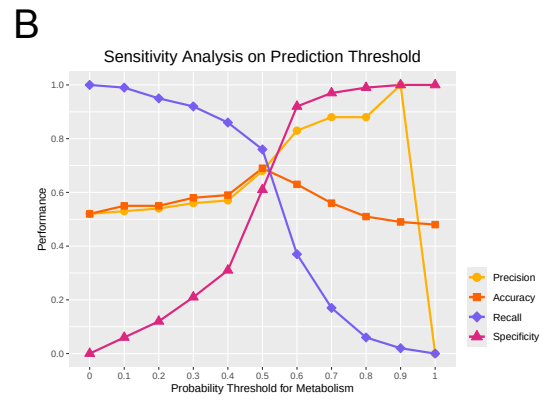


A

Confusion Matrices			
Metabolizing Status			Trained Models (n)
Truth	Predicted	Group	Multi-GutV1
Non-Met.	Non-Met.	True Neg.	21
Non-Met.	Met.	False Pos.	7
Met.	Non-Met.	False Neg.	7
Met.	Met.	True Pos.	24

Performance	
	Trained Models
Metric	Multi-GutV1
Precision	0.77
Accuracy	0.76
Recall	0.77
Specificity	0.75
ROC-AUC	0.83
PR-AUC	0.87



Supplementary 6: Model performance and results for Multi-GutV1. A) The confusion matrix and performance metrics are reported for the testing fold. B) Sensitivity analysis on threshold for prediction and its effect on performance. C) The sensitivity-specificity curve and the precision-recall curve determined from the test data displayed on the outer graph. The aggregated K-fold results are displayed in the inner graph. The MPPs grouped by microbes (D) and by drugs (E) determined from the testing and training data (left) and auxiliary predictions with unknown true class (right).